Домашнее задание („Обратная матрица")

№ 1.4.37.

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix}$$

1) $\det A = \begin{vmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{vmatrix} = (-1)\cdot 0\cdot 0 + 0\cdot 0{,}5\cdot 0 + 0\cdot 2\cdot 0 -$

$- 0\cdot 0\cdot 0 - 0{,}5\cdot 2\cdot(-1) - 0\cdot 0\cdot 2 = 1 \neq 0 \Rightarrow \exists A^{-1}$

2) $A_{11} = (-1)^{1+1}\cdot\begin{vmatrix} 0 & 2 \\ 0{,}5 & 0 \end{vmatrix} = 0\cdot 0 - 2\cdot 0{,}5 = -1$

$A_{12} = (-1)^{1+2}\cdot\begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix} = -(0\cdot 0 - 2\cdot 0) = 0$

$A_{13} = (-1)^{1+3}\cdot\begin{vmatrix} 0 & 0 \\ 0 & 0{,}5 \end{vmatrix} = 0\cdot 0{,}5 - 0\cdot 0 = 0$

$A_{21} = (-1)^{2+1}\cdot\begin{vmatrix} 0 & 0 \\ 0{,}5 & 0 \end{vmatrix} = -(0\cdot 0 - 0\cdot 0{,}5) = 0$

$A_{22} = (-1)^{2+2}\cdot\begin{vmatrix} -1 & 0 \\ 0 & 0 \end{vmatrix} = (-1)\cdot 0 - 0\cdot 0 = 0$

$A_{23} = (-1)^{2+3}\cdot\begin{vmatrix} -1 & 0 \\ 0 & 0{,}5 \end{vmatrix} = -((-1)\cdot 0{,}5 - 0\cdot 0) = 0{,}5$

$A_{31} = (-1)^{3+1}\cdot\begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0\cdot 2 - 0\cdot 0 = 0$

$A_{32} = (-1)^{3+2}\cdot\begin{vmatrix} -1 & 0 \\ 0 & 2 \end{vmatrix} = -((-1)\cdot 2 - 0\cdot 0) = 2$

$A_{33} = (-1)^{3+3}\cdot\begin{vmatrix} -1 & 0 \\ 0 & 0 \end{vmatrix} = (-1)\cdot 0 - 0\cdot 0 = 0$

3) $\tilde{A} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0{,}5 \\ 0 & 2 & 0 \end{pmatrix}^T = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix}$

4) $A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{1} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0{,}5 & 0 \end{pmatrix}$

№ 1.4.38.

$A = \begin{pmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{pmatrix}$

1) $\det A = \begin{vmatrix} 1 & 1 & -1 \\ 8 & 3 & -6 \\ -4 & -1 & 3 \end{vmatrix} = 1 \cdot \overset{9}{3 \cdot 3} \overset{+}{+} 1 \cdot \overset{24}{(-6)} \cdot (-4) \overset{+}{+} 8 \cdot \overset{8}{(-1)} \cdot (-1) -$

$- \overset{-12}{(-4)} \cdot 3 \cdot (-1) - \overset{-24}{8 \cdot 1 \cdot 3} - \overset{-6}{(-1)} \cdot (-6) \cdot 1 = -1 \neq 0 \Rightarrow \exists A^{-1}$

2) $A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 3 & -6 \\ -1 & 3 \end{vmatrix} = 3 \cdot 3 - (-6) \cdot (-1) = 3$

$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 8 & -6 \\ -4 & 3 \end{vmatrix} = -\left(\overset{24}{8 \cdot 3} - \overset{24}{(-6) \cdot (-4)}\right) = 0$

$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 8 & 3 \\ -4 & -1 \end{vmatrix} = \overset{-8}{8 \cdot (-1)} - \overset{12}{(-4)} \cdot 3 = 4$

$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = -\left(1 \cdot 3 - (-1) \cdot (-1)\right) = -2$

$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1 & -1 \\ -4 & 3 \end{vmatrix} = 1 \cdot 3 - (-4) \cdot (-1) = -1$

$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1 & 1 \\ -4 & -1 \end{vmatrix} = -\left(\overset{-1}{1 \cdot (-1)} \overset{+}{-} \overset{4}{1 \cdot (-4)}\right) = -3$

$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 1 & -1 \\ 3 & -6 \end{vmatrix} = 1 \cdot (-6) - (-1) \cdot 3 = 3$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & -1 \\ 8 & -6 \end{vmatrix} = -\left(1\cdot(-6) - (-1)\cdot 8\right) = -2$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 1 \\ 8 & 3 \end{vmatrix} = 1\cdot 3 - 1\cdot 8 = -5$$

3) $$\tilde{A} = \begin{pmatrix} 3 & 0 & 4 \\ -2 & -1 & -3 \\ 3 & -2 & -5 \end{pmatrix}^T = \begin{pmatrix} 3 & -2 & 3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix}$$

4) $$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{-1} \cdot \begin{pmatrix} 3 & -2 & 3 \\ 0 & -1 & -2 \\ 4 & -3 & -5 \end{pmatrix} =$$

$$= \begin{pmatrix} -3 & 2 & -3 \\ 0 & 1 & 2 \\ -4 & 3 & 5 \end{pmatrix}$$

~ 1.4.39

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{pmatrix}$$

1) $$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 2 \\ 4 & 1 & 4 \end{vmatrix} = \overset{-4}{1\cdot(-1)\cdot 4} \overset{+8}{+ 1\cdot 2\cdot 4} \overset{+4}{+ 2\cdot 1\cdot 2} -$$

$$\overset{+8}{- 4\cdot(-1)\cdot 2} \overset{-8}{- 2\cdot 1\cdot 4} \overset{-2}{- 1\cdot 1\cdot 2} = 6 \neq 0 \Rightarrow \exists A^{-1}$$

2) $$A_{11} = + \cdot \begin{vmatrix} -1 & 2 \\ 1 & 4 \end{vmatrix} = -1\cdot 4 - 2\cdot 1 = -6$$

$$A_{12} = - \cdot \begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix} = -\left(2\cdot 4 - 2\cdot 4\right) = 0$$

$$A_{13} = + \cdot \begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix} = 2\cdot 1 - (-1)\cdot 4 = 6$$

$$A_{21} = - \cdot \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -\left(1\cdot 4 - 2\cdot 1\right) = -2$$

$$A_{22} = + \cdot \begin{vmatrix} 1 & 2 \\ 4 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 4 = -4$$

$$A_{23} = - \cdot \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = -(1 \cdot 1 - 1 \cdot 4) = 3$$

$$A_{31} = + \cdot \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = 1 \cdot 2 - (-1) \cdot 2 = 4$$

$$A_{32} = - \cdot \begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} = -(1 \cdot 2 - 2 \cdot 2) = 2$$

$$A_{33} = + \cdot \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = 1 \cdot (-1) - 1 \cdot 2 = -3$$

3) $$\tilde{A} = \begin{pmatrix} -6 & 0 & 6 \\ -2 & -4 & 3 \\ 4 & 2 & -3 \end{pmatrix}^T = \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix}$$

4) $$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{6} \cdot \begin{pmatrix} -6 & -2 & 4 \\ 0 & -4 & 2 \\ 6 & 3 & -3 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{1}{6} \cdot (-6) & \frac{1}{6} \cdot (-2) & \frac{1}{6} \cdot 4 \\ \frac{1}{6} \cdot 0 & \frac{1}{6} \cdot (-4) & \frac{1}{6} \cdot 2 \\ \frac{1}{6} \cdot 6 & \frac{1}{6} \cdot 3 & \frac{1}{6} \cdot (-3) \end{pmatrix} = \begin{pmatrix} -1 & -\frac{1}{3} & \frac{2}{3} \\ 0 & -\frac{2}{3} & \frac{1}{3} \\ 1 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

2.4.40

$$A = \begin{pmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ 1 & 5 & 1 \end{pmatrix}$$

1) $$\det A = \begin{vmatrix} 3 & 4 & 2 \\ 2 & -4 & -3 \\ 1 & 5 & 1 \end{vmatrix} = 3 \cdot \overset{-12}{(-4)} \cdot 1 \overset{-}{+} 4 \cdot \overset{12}{(-3)} \cdot 1 \overset{+}{+} 2 \cdot \overset{20}{5} \cdot 2 -$$

$$\overset{+8}{-1 \cdot (-4) \cdot 2} \overset{-8}{-} 2 \cdot 4 \cdot 1 \overset{+45}{-} 5 \cdot (-3) \cdot 3 = 41 \neq 0 \Rightarrow \exists A^{-1}$$

2) $A_{11} = + \cdot \begin{vmatrix} -4 & -3 \\ 5 & 1 \end{vmatrix} = (-4)\cdot 1 - (-3)\cdot 5 = 11$

$A_{12} = - \cdot \begin{vmatrix} 2 & -3 \\ 1 & 1 \end{vmatrix} = -(2\cdot 1 - (-3)\cdot 1) = -5$

$A_{13} = + \cdot \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = 2\cdot 5 - (-4)\cdot 1 = 14$

$A_{21} = - \cdot \begin{vmatrix} 4 & 2 \\ 5 & 1 \end{vmatrix} = -(4\cdot 1 - 5\cdot 2) = 6$

$A_{22} = + \cdot \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = 3\cdot 1 - 2\cdot 1 = 1$

$A_{23} = - \cdot \begin{vmatrix} 3 & 4 \\ 1 & 5 \end{vmatrix} = -(3\cdot 5 - 4\cdot 1) = -11$

$A_{31} = + \cdot \begin{vmatrix} 4 & 2 \\ -4 & -3 \end{vmatrix} = 4\cdot(-3) - 2\cdot(-4) = -4$

$A_{32} = - \cdot \begin{vmatrix} 3 & 2 \\ 2 & -3 \end{vmatrix} = -(3\cdot(-3) - 2\cdot 2) = 13$

$A_{33} = + \cdot \begin{vmatrix} 3 & 4 \\ 2 & -4 \end{vmatrix} = 3\cdot(-4) - 4\cdot 2 = -20$

3) $\tilde{A} = \begin{pmatrix} 11 & -5 & 14 \\ 6 & 1 & -11 \\ -4 & 13 & -20 \end{pmatrix}^T = \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix}$

4) $A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{41} \cdot \begin{pmatrix} 11 & 6 & -4 \\ -5 & 1 & 13 \\ 14 & -11 & -20 \end{pmatrix} =$

$= \begin{pmatrix} \frac{1}{41}\cdot 11 & \frac{1}{41}\cdot 6 & \frac{1}{41}\cdot(-4) \\ \frac{1}{41}\cdot(-5) & \frac{1}{41}\cdot 1 & \frac{1}{41}\cdot 13 \\ \frac{1}{41}\cdot 14 & \frac{1}{41}\cdot(-11) & \frac{1}{41}\cdot(-20) \end{pmatrix} = \begin{pmatrix} \frac{11}{41} & \frac{6}{41} & -\frac{4}{41} \\ -\frac{5}{41} & \frac{1}{41} & \frac{13}{41} \\ \frac{14}{41} & -\frac{11}{41} & -\frac{20}{41} \end{pmatrix}$

№ 1.4.41.

$$A = \begin{pmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{pmatrix}$$

1) $\det A = \begin{vmatrix} 3 & -1 & 2 \\ 4 & -3 & 3 \\ 1 & 3 & 0 \end{vmatrix} = 3\cdot\overset{0}{(-3)}\cdot 0 + 4\cdot\overset{24}{3}\cdot 2 + \overset{-3}{(-1)}\cdot 3\cdot 1 -$

$- \overset{+6}{1\cdot(-3)\cdot 2} - 4\cdot\overset{0}{(-1)}\cdot 0 - \overset{-27}{3\cdot 3\cdot 3} = 0 \Rightarrow$ обратной матрицы A^{-1} не существует.

№ 1.4.42.

$$A = \begin{pmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

1) $\det A = \begin{vmatrix} 5 & 8 & -1 \\ 2 & -3 & 2 \\ 1 & 2 & 3 \end{vmatrix} = \overset{-45}{5\cdot(-3)\cdot 3} + \overset{+16}{8\cdot 2\cdot 1} + \overset{-4}{2\cdot 2\cdot(-1)} - \overset{-3}{1\cdot(-3)\cdot(-1)} -$

$- \overset{-48}{8\cdot 2\cdot 3} - \overset{-20}{2\cdot 2\cdot 5} = -104$

2) $A_{11} = + \cdot \begin{vmatrix} -3 & 2 \\ 2 & 3 \end{vmatrix} = -3\cdot 3 - 2\cdot 2 = -13$

$A_{12} = - \cdot \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} = -(2\cdot 3 - 2\cdot 1) = -4$

$A_{13} = + \cdot \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2\cdot 2 - (-3)\cdot 1 = 7$

$A_{21} = - \cdot \begin{vmatrix} 8 & -1 \\ 2 & 3 \end{vmatrix} = -(8\cdot 3 - (-1)\cdot 2) = -26$

$A_{22} = + \cdot \begin{vmatrix} 5 & -1 \\ 1 & 3 \end{vmatrix} = 5\cdot 3 - (-1)\cdot 1 = 16$

$A_{23} = - \cdot \begin{vmatrix} 5 & 8 \\ 1 & 2 \end{vmatrix} = -(5\cdot 2 - 8\cdot 1) = -2$

$$A_{31} = + \cdot \begin{vmatrix} 8 & -1 \\ -3 & 2 \end{vmatrix} = 8 \cdot 2 - (-1) \cdot (-3) = 13$$

$$A_{32} = - \cdot \begin{vmatrix} 5 & -1 \\ 2 & 2 \end{vmatrix} = -(5 \cdot 2 - (-1) \cdot 2) = -12$$

$$A_{33} = + \cdot \begin{vmatrix} 5 & 8 \\ 2 & -3 \end{vmatrix} = 5 \cdot (-3) - 8 \cdot 2 = -31$$

3) $$\tilde{A} = \begin{pmatrix} -13 & -4 & 7 \\ -26 & 16 & -2 \\ 13 & -12 & -31 \end{pmatrix}^T = \begin{pmatrix} -13 & -26 & 13 \\ -4 & 16 & -12 \\ 7 & -2 & -31 \end{pmatrix}$$

4) $$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = -\frac{1}{104} \cdot \begin{pmatrix} -13 & -26 & 13 \\ -4 & 16 & -12 \\ 7 & -2 & -31 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{104} \cdot (-13) & -\frac{1}{104} \cdot (-26) & -\frac{1}{104} \cdot 13 \\ -\frac{1}{104} \cdot (-4) & -\frac{1}{104} \cdot 16 & -\frac{1}{104} \cdot (-12) \\ -\frac{1}{104} \cdot 7 & -\frac{1}{104} \cdot (-2) & -\frac{1}{104} \cdot (-31) \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{1}{8} & \frac{1}{4} & -\frac{1}{8} \\ \frac{1}{26} & -\frac{2}{13} & \frac{3}{26} \\ -\frac{7}{104} & \frac{1}{52} & \frac{31}{104} \end{pmatrix}$$

∞ 1.4.43.

$$A = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$$

1) $$\det A = \begin{vmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 1 \cdot 2 \cdot 2 + (-1) \cdot 1 \cdot (-1) +$$

$$+ (-1) \cdot (-1) \cdot 1 - (-1) \cdot 2 \cdot (-1) - (-1) \cdot (-1) \cdot 2 - 1 \cdot 1 \cdot (1) = 1 \neq 0 \Rightarrow \exists A^{-1}$$

2) $$\Gamma = (A \mid E) = \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ \mathrm{II} + \mathrm{I} \\ \mathrm{III} + \mathrm{I} \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} \mathrm{I} + \mathrm{III} \\ \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} \mathrm{I} + \mathrm{II} \\ \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right) = \left(E \mid A^{-1}\right) = \Gamma_2$$

Torga:

$$A^{-1} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

2) 1.4.44.

$$A = \begin{pmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 3 \end{pmatrix}$$

1) $\det A = \begin{vmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 3 \end{vmatrix} = \overset{54}{2 \cdot 9 \cdot 3} \overset{+}{+} \overset{28}{7 \cdot 4 \cdot 1} \overset{+}{+} \overset{45}{3 \cdot 5 \cdot 3} - \overset{-27}{1 \cdot 9 \cdot 3} \overset{-63}{-} \overset{-40}{3 \cdot 3 \cdot 7} - 5 \cdot 4 \cdot 2 = -3 \neq 0 \Rightarrow \exists A^{-1}$

2) $$\Gamma = (A \mid E) = \left(\begin{array}{ccc|ccc} 2 & 7 & 3 & 1 & 0 & 0 \\ 3 & 9 & 4 & 0 & 1 & 0 \\ 1 & 5 & 3 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ 2 \cdot \mathrm{II} - 3 \cdot \mathrm{I} \\ 2 \cdot \mathrm{III} - \mathrm{I} \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 2 & 7 & 3 & 1 & 0 & 0 \\ 0 & -3 & -1 & -3 & 2 & 0 \\ 0 & 3 & 3 & -1 & 0 & 2 \end{array}\right) \begin{array}{l} \\ \\ \mathrm{III} + \mathrm{II} \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 2 & 7 & 3 & 1 & 0 & 0 \\ 0 & -3 & -1 & -3 & 2 & 0 \\ 0 & 0 & 2 & -4 & 2 & 2 \end{array}\right) \begin{array}{l} \mathrm{I} : 2 \\ \\ \mathrm{III} : 2 \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & -3 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \quad \text{II} : (-3) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \quad \frac{2}{7} \cdot \text{I} - \text{II} \quad \sim$$

$$\sim \left(\begin{array}{ccc|ccc} \frac{2}{7} & 0 & \frac{2}{21} & -\frac{6}{7} & \frac{2}{3} & 0 \\ 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \quad \frac{7}{2} \cdot \text{I} \quad \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & -3 & \frac{7}{3} & 0 \\ 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \quad \text{II} - \frac{1}{3} \cdot \text{III} \quad \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & -3 & \frac{7}{3} & 0 \\ 0 & 1 & 0 & 5/3 & -1 & -1/3 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) \quad \text{I} - \frac{1}{3} \cdot \text{III} \quad \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{7}{3} & 2 & -\frac{1}{3} \\ 0 & 1 & 0 & \frac{5}{3} & -1 & -\frac{1}{3} \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right) = (E \mid A^{-1}) = \Gamma_2 .$$

Torga,

$$A^{-1} = \begin{pmatrix} -\frac{7}{3} & 2 & -\frac{1}{3} \\ \frac{5}{3} & -1 & -\frac{1}{3} \\ -2 & 1 & 1 \end{pmatrix}$$

№ 1.4.45.

$$A = \begin{pmatrix} 1 & 2 & -2 & 4 \\ 2 & 6 & 1 & 0 \\ 3 & 0 & 1 & 2 \\ -1 & 4 & 5 & -4 \end{pmatrix}$$

1) $$\det A = \begin{vmatrix} \overset{+}{1} & \overset{-}{2} & \overset{+}{-2} & \overset{-}{4} \\ \overset{-}{2} & \overset{+}{6} & \overset{-}{1} & \overset{+}{0} \\ \overset{+}{3} & \overset{-}{0} & \overset{+}{1} & \overset{-}{2} \\ \overset{-}{-1} & \overset{+}{4} & \overset{-}{5} & \overset{+}{-4} \end{vmatrix} = -2 \cdot \begin{vmatrix} 2 & -2 & 4 \\ 0 & 1 & 2 \\ 4 & 5 & -4 \end{vmatrix} + 6 \cdot \begin{vmatrix} 1 & -2 & 4 \\ 3 & 1 & 2 \\ -1 & 5 & -4 \end{vmatrix} -$$

$$-1 \cdot \begin{vmatrix} 1 & 2 & 4 \\ 3 & 0 & 2 \\ -1 & 4 & -4 \end{vmatrix} + 0 \cdot |\ldots| = -2(\overset{-8}{2 \cdot 1 \cdot (-4)} + \overset{+0}{5 \cdot 0 \cdot 4} + \overset{-16}{(-2) \cdot 2 \cdot 4} - \overset{-16}{4 \cdot 1 \cdot 4} -$$

$$- \overset{-0}{0 \cdot (-2) \cdot (-4)} - \overset{-20}{5 \cdot 2 \cdot 2}) + 6(\overset{-4}{1 \cdot 1 \cdot (-4)} + \overset{+4}{(-2) \cdot 2 \cdot (-1)} + \overset{+60}{3 \cdot 5 \cdot 4} - \overset{+4}{(-1) \cdot 1 \cdot 4} -$$

$$- \overset{-24}{3 \cdot (-2) \cdot (-4)} - \overset{-10}{5 \cdot 2 \cdot 1}) - 1(\overset{0}{1 \cdot 0 \cdot (-4)} + \overset{-4}{2 \cdot 2 \cdot (-1)} + \overset{+48}{3 \cdot 4 \cdot 4} - \overset{+0}{(-1) \cdot 0 \cdot 4} -$$

$$+ \overset{24}{3 \cdot 2 \cdot (-4)} - \overset{-8}{4 \cdot 2 \cdot 1}) + 0 \cdot |\ldots| = -2 \cdot (-60) + 6 \cdot 30 - 1 \cdot 60 = 240 \neq 0 \Rightarrow \exists A^{-1}$$

$$\Gamma = (A \mid E) = \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 2 & 6 & 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ -1 & 4 & 5 & -4 & 0 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ \underline{II} - 2 \cdot I \\ \underline{III} - 3 \cdot I \\ \underline{IV} + I \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 2 & 5 & -8 & -2 & 1 & 0 & 0 \\ 0 & -6 & 7 & -10 & -3 & 0 & 1 & 0 \\ 0 & 6 & 3 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \\ \\ \underline{III} + 3 \cdot \underline{II} \\ \underline{IV} - 3 \cdot \underline{II} \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 2 & 5 & -8 & -2 & 1 & 0 & 0 \\ 0 & 0 & 22 & -34 & -9 & 3 & 1 & 0 \\ 0 & 0 & -12 & 24 & 7 & -3 & 0 & 1 \end{array}\right) \begin{array}{l} \\ \underline{II} : 2 \\ \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 2 & -2 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 22 & -34 & -9 & 3 & 1 & 0 \\ 0 & 0 & -12 & 24 & 7 & -3 & 0 & 1 \end{array}\right) \begin{array}{l} I - 2 \cdot \underline{II} \\ \\ \\ 11 \cdot \underline{IV} + 6 \cdot \underline{III} \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 22 & -34 & -9 & 3 & 1 & 0 \\ 0 & 0 & 0 & 60 & 23 & -15 & 6 & 11 \end{array}\right) \begin{array}{l} \\ \\ \underline{III} : 22 \\ \underline{IV} : 60 \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & -7 & 12 & 3 & -1 & 0 & 0 \\ 0 & 1 & 2{,}5 & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 1 & -\frac{17}{11} & -\frac{9}{22} & \frac{3}{22} & \frac{1}{22} & 0 \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right) \begin{array}{l} \mathrm{I} + 7 \cdot \mathrm{III} \\ \\ \sim \\ \\ \end{array}$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & \frac{13}{11} & \frac{3}{22} & -\frac{1}{22} & \frac{7}{22} & 0 \\ 0 & 1 & \frac{5}{2} & -4 & -1 & 0{,}5 & 0 & 0 \\ 0 & 0 & 1 & -\frac{17}{11} & -\frac{9}{22} & \frac{3}{22} & \frac{1}{22} & 0 \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right) \begin{array}{l} \mathrm{I} - \frac{13}{11} \cdot \mathrm{IV} \\ \mathrm{II} + 4 \cdot \mathrm{IV} \\ \mathrm{III} + \frac{17}{11} \cdot \mathrm{IV} \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{19}{60} & \frac{1}{4} & \frac{1}{5} & -\frac{13}{60} \\ 0 & 1 & \frac{5}{2} & 0 & \frac{8}{15} & -\frac{1}{2} & \frac{2}{5} & \frac{11}{15} \\ 0 & 0 & 1 & 0 & \frac{11}{60} & -\frac{1}{4} & \frac{1}{5} & \frac{17}{60} \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right) \begin{array}{l} \\ \mathrm{II} - \frac{5}{2} \cdot \mathrm{III} \\ \\ \\ \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{19}{60} & \frac{1}{4} & \frac{1}{5} & -\frac{13}{60} \\ 0 & 1 & 0 & 0 & \frac{3}{40} & \frac{1}{8} & -\frac{1}{10} & \frac{1}{40} \\ 0 & 0 & 1 & 0 & \frac{11}{60} & -\frac{1}{4} & \frac{1}{5} & \frac{17}{60} \\ 0 & 0 & 0 & 1 & \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{array}\right) = \left(E \mid A^{-1}\right) = \Gamma_2$$

$$A^{-1} = \begin{pmatrix} -\frac{19}{60} & \frac{1}{4} & \frac{1}{5} & -\frac{13}{60} \\ \frac{3}{40} & \frac{1}{8} & -\frac{1}{10} & \frac{1}{40} \\ \frac{11}{60} & -\frac{1}{4} & \frac{1}{5} & \frac{17}{60} \\ \frac{23}{60} & -\frac{15}{60} & \frac{1}{10} & \frac{11}{60} \end{pmatrix}$$

№ 1.4.50.

$$X \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$X \cdot A = B \Rightarrow X = B \cdot A^{-1}$$

1) $\det A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0 \Rightarrow \exists A^{-1}$

2) $A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{\det A} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$

$$= \frac{1}{-2} \cdot \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} \left(-\frac{1}{2}\right) \cdot 4 & \left(-\frac{1}{2}\right) \cdot (-2) \\ \left(-\frac{1}{2}\right) \cdot (-3) & \left(-\frac{1}{2}\right) \cdot 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

3) $X = B \cdot A^{-1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} =$

$= \begin{pmatrix} 0\cdot(-2)+0\cdot 1{,}5 & 0\cdot 2+0\cdot 1{,}5 \\ 0\cdot 1+0\cdot(-0{,}5) & 0\cdot 1+0\cdot(-0{,}5) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

№ 1.4.51.

$X \cdot \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$XA = B \Rightarrow X = B \cdot A^{-1}$

1) $\det A = \begin{vmatrix} 4 & 3 \\ -5 & -4 \end{vmatrix} = 4\cdot(-4) - 3\cdot(-5) = -1 \neq 0 \Rightarrow \exists A^{-1}$

2) $A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{\det A} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$

$= \frac{1}{-1} \cdot \begin{pmatrix} -4 & -3 \\ 5 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}$

3) $X = B \cdot A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix} =$

$= \begin{pmatrix} \overset{4}{1\cdot 4+0\cdot(-5)} & 0\cdot(-4)+1\cdot 3 \\ 1\cdot(-5)+0\cdot 4 & 0\cdot 3+\overset{-4}{1\cdot(-4)} \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}$

№ 1.4.52.

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$A \cdot X = B \Rightarrow X = A^{-1} \cdot B$

1) $\det A = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1\cdot 1 - 1\cdot 1 = 0 \Rightarrow \nexists A^{-1}$

Ответ: не существует X.

№ 1.4.53.

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$AX = B \Rightarrow X = A^{-1} \cdot B$

1) $\det A = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1\cdot 1 - 1\cdot 1 = 0 \Rightarrow \nexists A^{-1}$

Ответ: не существует X.

№ 1 4.54.

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot X \cdot \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$$AXC = B \Rightarrow X = A^{-1}\cdot B \cdot C^{-1}$$

1) $\det A = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = \overset{3}{1\cdot 3} \overset{+}{-} \overset{2}{(-1)\cdot 2} = 5 \neq 0 \Rightarrow \exists A^{-1}$

2) $\det C = \begin{vmatrix} -5 & 6 \\ -4 & 5 \end{vmatrix} = \overset{-25}{-5\cdot 5} \overset{+}{-} \overset{24}{(-4)\cdot 6} = -1 \neq 0 \Rightarrow \exists C^{-1}$

3) $A^{-1} = \dfrac{1}{\det A}\cdot \tilde{A} = \dfrac{1}{\det A}\cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$

$$= \frac{1}{5}\cdot\begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{5}\cdot 3 & \frac{1}{5}\cdot 1 \\ \frac{1}{5}\cdot(-2) & \frac{1}{5}\cdot 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix}$$

4) $C^{-1} = \dfrac{1}{\det C}\cdot \tilde{C} = \dfrac{1}{\det C}\cdot \begin{pmatrix} c_{22} & -c_{12} \\ -c_{21} & c_{11} \end{pmatrix} =$

$$= \frac{1}{-1}\cdot\begin{pmatrix} 5 & -6 \\ 4 & -5 \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix}$$

5) $X = A^{-1}\cdot B\cdot C^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix}\cdot\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}\cdot\begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} =$

$$= \begin{pmatrix} \frac{3}{5}\cdot 1 + \frac{1}{5}\cdot 2 & \frac{3}{5}\cdot(-1) + \frac{1}{5}\cdot 3 \\ -\frac{2}{5}\cdot 1 + \frac{1}{5}\cdot 2 & -\frac{2}{5}\cdot(-1) + \frac{1}{5}\cdot 3 \end{pmatrix}\cdot\begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\cdot\begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1\cdot(-5) + 0\cdot(-4) & 1\cdot 6 + 0\cdot 5 \\ 0\cdot(-5) + 1\cdot(-4) & 0\cdot 6 + 1\cdot 5 \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ -4 & 5 \end{pmatrix}$$

№ 2.4.55.

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot X \cdot \begin{pmatrix} 2 & -2 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$$AXC = B \Rightarrow X = A^{-1} \cdot B \cdot C^{-1}$$

1) $\det A = \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = \overset{3}{1 \cdot 3} - \overset{-2}{(-1) \cdot 2} = 5 \neq 0 \Rightarrow \exists A^{-1}$

2) $\det C = \begin{vmatrix} 2 & -2 \\ -4 & 5 \end{vmatrix} = \overset{10}{2 \cdot 5} - \overset{8}{(-2)(-4)} = 2 \neq 0 \Rightarrow \exists C^{-1}$

3) $A^{-1} = \dfrac{1}{\det A} \cdot \tilde{A} = \dfrac{1}{\det A} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} =$

$$= \frac{1}{5} \cdot \begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \cdot 3 & 1 \cdot \frac{1}{5} \\ (-2) \cdot \frac{1}{5} & 1 \cdot \frac{1}{5} \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix}$$

4) $C^{-1} = \dfrac{1}{\det C} \cdot \tilde{C} = \dfrac{1}{\det C} \cdot \begin{pmatrix} C_{22} & -C_{12} \\ -C_{21} & C_{11} \end{pmatrix} =$

$$= \frac{1}{2} \begin{pmatrix} 5 & 2 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \cdot 5 & \frac{1}{2} \cdot 2 \\ \frac{1}{2} \cdot 4 & \frac{1}{2} \cdot 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix}$$

5) $X = A^{-1} \cdot B \cdot C^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix} =$

$$= \begin{pmatrix} \frac{3}{5} \cdot 1 + \frac{1}{5} \cdot 2 & \frac{3}{5} \cdot (-1) + \frac{1}{5} \cdot 3 \\ -\frac{2}{5} \cdot 1 + \frac{1}{5} \cdot 2 & -\frac{2}{5} \cdot (-1) + \frac{1}{5} \cdot 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot \frac{5}{2} + 0 \cdot 2 & 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot \frac{5}{2} + 1 \cdot 2 & 0 \cdot 1 + 1 \cdot 1 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{5}{2} & 1 \\ 2 & 1 \end{pmatrix}$$

№ 1.4.56.

$$X \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$XA = B \Rightarrow X = B \cdot A^{-1}$

1) $\det A = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 1 \cdot 2 \cdot 3 + 0 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot 0 - 0 \cdot 2 \cdot 0 - 0 \cdot 0 \cdot 1 - 0 \cdot 0 \cdot 3 =$

$= 6 \neq 0 \Rightarrow \exists A^{-1}$

2) $A_{11} = +\begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 2 \cdot 3 - 0 \cdot 0 = 6$

$A_{12} = -\begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = 0 \cdot 3 - 0 \cdot 0 = 0$

$A_{13} = +\begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix} = 0 \cdot 0 - 2 \cdot 0 = 0$

$A_{21} = -\begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = 0 \cdot 3 - 0 \cdot 0 = 0$

$A_{22} = +\begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} = 1 \cdot 3 - 0 \cdot 0 = 3$

$A_{23} = -\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 1 \cdot 0 - 0 \cdot 0 = 0$

$A_{31} = +\begin{vmatrix} 0 & 0 \\ 2 & 0 \end{vmatrix} = 0 \cdot 0 - 0 \cdot 2 = 0$

$A_{32} = -\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 1 \cdot 0 - 0 \cdot 0 = 0$

$A_{33} = +\begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 1 \cdot 2 - 0 \cdot 0 = 2$

2)

$$\tilde{A} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}^T = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \tilde{A} = \frac{1}{6} \cdot \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{1}{6} \cdot 6 & \frac{1}{6} \cdot 0 & \frac{1}{6} \cdot 0 \\ \frac{1}{6} \cdot 0 & \frac{1}{6} \cdot 3 & \frac{1}{6} \cdot 0 \\ \frac{1}{6} \cdot 0 & \frac{1}{6} \cdot 0 & \frac{1}{6} \cdot 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0{,}5 & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$$

3) $$X = B \cdot A^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0{,}5 & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} =$$

$$= \begin{pmatrix} 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 & 0 \cdot 0 + 0 \cdot 0{,}5 + 1 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 + 1 \cdot \frac{1}{3} \\ 0 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 & 0 \cdot 0 + 2 \cdot 0{,}5 + 0 \cdot 0 & 0 \cdot 0 + 2 \cdot 0 + 0 \cdot 1/3 \\ 3 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 & 3 \cdot 0 + 0 \cdot 0{,}5 + 0 \cdot 0 & 3 \cdot 0 + 0 \cdot 0 + 0 \cdot 1/3 \end{pmatrix} =$$

$$= \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 1 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

№ 1.4.57.

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \qquad AX = B \Rightarrow X = A^{-1} \cdot B$$

1) $$\det A = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{vmatrix} = 1 \cdot \overset{3}{3} \cdot 1 + (-2) \cdot \overset{0}{(-1)} \cdot 0 + 2 \cdot \overset{12}{(-2)} \cdot 3 -$$

$$- \overset{0}{0} \cdot 3 \cdot 3 - 2 \cdot \overset{4}{(-2)} \cdot 1 - \overset{2}{(-2)} \cdot (-1) \cdot 1 = -7 \neq 0 \Rightarrow \exists A^{-1}$$

2) $A_{11} = \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} = 3\cdot 1 - (-1)\cdot(-2) = 1$

$A_{12} = -\begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = -(2\cdot 1 - (-1)\cdot 0) = -2$

$A_{13} = \begin{vmatrix} 2 & 3 \\ 0 & -2 \end{vmatrix} = 2\cdot(-2) - 3\cdot 0 = -4$

$A_{21} = -\begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} = -(\overset{2}{(-2)\cdot 1} \overset{+}{-} \overset{6}{3\cdot(-2)}) = -4$

$A_{22} = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1\cdot 1 - 3\cdot 0 = 1$

$A_{23} = -\begin{vmatrix} 1 & -2 \\ 0 & -2 \end{vmatrix} = -(-2)\cdot 1 - (-2)\cdot 0 = 2$

$A_{31} = \begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = \overset{2}{-2(-1)} \overset{-}{-} \overset{9}{3\cdot 3} = -7$

$A_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -(\overset{-1}{1\cdot(-1)} \overset{-}{-} \overset{6}{3\cdot 2}) = 7$

$A_{33} = \begin{vmatrix} 1 & -2 \\ 2 & 3 \end{vmatrix} = \overset{3}{1\cdot 3} \overset{+}{-} \overset{4}{(-2)\cdot 2} = 7$

$$\tilde{A} = \begin{pmatrix} 1 & -2 & -4 \\ -4 & 1 & 2 \\ -7 & 7 & 7 \end{pmatrix}^T = \begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A}\cdot\tilde{A} = \frac{1}{-7}\cdot\begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{7}\cdot 1 & -\frac{1}{7}(-4) & -\frac{1}{7}\cdot(-7) \\ -\frac{1}{7}(-2) & -\frac{1}{7}\cdot 1 & -\frac{1}{7}\cdot(7) \\ -\frac{1}{7}\cdot(-4) & -\frac{1}{7}\cdot(2) & -\frac{1}{7}\cdot 7 \end{pmatrix} = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}$$

3) $X = A^{-1} \cdot B = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} =$

$$= \begin{pmatrix} -\frac{1}{7} \cdot 2 + \frac{4}{7} \cdot (-1) + 1 \cdot 3 \\ \frac{2}{7} \cdot 2 + \left(-\frac{1}{7}\right) \cdot (-1) + (-1) \cdot 3 \\ \frac{4}{7} \cdot 2 + \left(-\frac{2}{7}\right) \cdot (-1) + (-1) \cdot 3 \end{pmatrix} = \begin{pmatrix} \frac{15}{7} \\ -\frac{16}{7} \\ -\frac{11}{7} \end{pmatrix}$$

№ 1.4.58.

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{pmatrix} \cdot X \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$$

$AXC = B \Rightarrow X = A^{-1} \cdot B \cdot C^{-1}$

1) $\det A = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 0 & -2 & 1 \end{vmatrix} = \overset{3}{1 \cdot 3 \cdot 1} + \overset{0}{(-2) \cdot (-1) \cdot 0} + \overset{-12}{2 \cdot (-2) \cdot 3} -$

$- \overset{0}{0 \cdot 3 \cdot 3} - \overset{4}{2 \cdot (-2) \cdot 1} - \overset{2}{(-2) \cdot (-1) \cdot 1} = -7 \neq 0 \Rightarrow \exists A^{-1}$

2) $\det C = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix} = \overset{0}{1 \cdot 5 \cdot 0} + \overset{84}{2 \cdot 6 \cdot 7} + \overset{96}{4 \cdot 8 \cdot 3} - \overset{105}{7 \cdot 5 \cdot 3} -$

$- \overset{0}{4 \cdot 2 \cdot 0} - \overset{48}{8 \cdot 6 \cdot 1} = 27 \neq 0 \Rightarrow \exists C^{-1}$

3) $A_{11} = \underset{+}{} \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} = 3 \cdot 1 - (-1) \cdot (-2) = 1$

$A_{12} = \underset{-}{} \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = -(2 \cdot 1 - (-1) \cdot 0) = -2$

$A_{13} = \underset{+}{} \begin{vmatrix} 2 & 3 \\ 0 & -2 \end{vmatrix} = 2 \cdot (-2) \cdot 3 \cdot 0 = -4$

$$A_{21} = -\begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} = -\left(\overset{+2}{(-2)\cdot 1} \overset{+6}{- 3\cdot(-2)}\right) = -4$$

$$A_{22} = +\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1\cdot 1 - 3\cdot 0 = 1$$

$$A_{23} = -\begin{vmatrix} 1 & -2 \\ 0 & -2 \end{vmatrix} = -\left(\overset{-2}{1\cdot(-2)} \overset{-}{-(-2)\cdot 0}\right) = 2$$

$$A_{31} = +\begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = (-2)\cdot(-1) - 3\cdot 3 = -7$$

$$A_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -\left(1\cdot(-1) - 3\cdot 2\right) = 7$$

$$A_{33} = +\begin{vmatrix} 1 & -2 \\ 2 & 3 \end{vmatrix} = 1\cdot 3 - (-2)\cdot 2 = 7$$

$$\tilde{A} = \begin{pmatrix} 1 & -2 & -4 \\ -4 & 1 & 2 \\ -7 & 7 & 7 \end{pmatrix}^T = \begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A}\cdot\tilde{A} = -\frac{1}{7}\cdot\begin{pmatrix} 1 & -4 & -7 \\ -2 & 1 & 7 \\ -4 & 2 & 7 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{7}\cdot 1 & -\frac{1}{7}\cdot(-4) & -\frac{1}{7}\cdot(-7) \\ -\frac{1}{7}\cdot(-2) & -\frac{1}{7}\cdot 1 & -\frac{1}{7}\cdot 7 \\ -\frac{1}{7}\cdot(-4) & -\frac{1}{7}\cdot 2 & -\frac{1}{7}\cdot 7 \end{pmatrix} = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}$$

4) $$A_{11} = \begin{vmatrix} 5 & 6 \\ 8 & 0 \end{vmatrix} = 5\cdot 0 - 6\cdot 8 = -48$$

$$A_{12} = -\begin{vmatrix} 4 & 6 \\ 7 & 0 \end{vmatrix} = -\left(4\cdot 0 - 6\cdot 7\right) = 42$$

$$A_{13} = \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = 4\cdot 8 - 5\cdot 7 = -3$$

$$A_{21} = -\begin{vmatrix} 2 & 3 \\ 8 & 0 \end{vmatrix} = -(2\cdot 0 - 3\cdot 8) = 24$$

$$A_{22} = \begin{vmatrix} 1 & 3 \\ 7 & 0 \end{vmatrix} = 1\cdot 0 - 3\cdot 7 = -21$$

$$A_{23} = -\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = -(1\cdot 8 - 2\cdot 7) = 6$$

$$A_{31} = \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 2\cdot 6 - 3\cdot 5 = -3$$

$$A_{32} = -\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = -(1\cdot 6 - 3\cdot 4) = 6$$

$$A_{33} = \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = 1\cdot 5 - 2\cdot 4 = -3$$

$$\tilde{C} = \begin{pmatrix} -48 & 42 & -3 \\ 24 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}^T = \begin{pmatrix} -48 & 24 & -3 \\ 42 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix}$$

$$C^{-1} = \frac{1}{\det C}\cdot \tilde{C} = \frac{1}{27}\cdot\begin{pmatrix} -48 & 24 & -3 \\ 42 & -21 & 6 \\ -3 & 6 & -3 \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{1}{27}\cdot(-48) & \frac{1}{27}\cdot 24 & \frac{1}{27}\cdot(-3) \\ \frac{1}{27}\cdot 42 & \frac{1}{27}\cdot(-21) & \frac{1}{27}\cdot 6 \\ \frac{1}{27}\cdot(-3) & \frac{1}{27}\cdot 6 & \frac{1}{27}\cdot(-3) \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix}$$

5) $X = A^{-1} \cdot B \cdot C^{-1} =$

$$= \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} \cdot \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{7}\cdot 1 + \frac{4}{7}\cdot 4 + 1\cdot 7 & -\frac{1}{7}\cdot 2 + \frac{4}{7}\cdot 5 + 1\cdot 8 & -\frac{1}{7}\cdot 3 + \frac{4}{7}\cdot 6 + 1\cdot 0 \\ \frac{2}{7}\cdot 1 + \left(-\frac{1}{7}\right)\cdot 4 + (-1)\cdot 7 & \frac{2}{7}\cdot 2 + \left(-\frac{1}{7}\right)\cdot 5\cdot(-1)\cdot 8 & \frac{2}{7}\cdot 3 + \left(-\frac{1}{7}\right)\cdot 6 + (-1)\cdot 0 \\ \frac{4}{7}\cdot 1 + \left(-\frac{2}{7}\right)\cdot 4 + (-1)\cdot 7 & 2\cdot\frac{4}{7} + \left(-\frac{2}{7}\right)\cdot 5 + (-1)\cdot 8 & \frac{4}{7}\cdot 3 + \left(-\frac{2}{7}\right)\cdot 6 + (-1)\cdot 0 \end{pmatrix} \cdot$$

$$\cdot \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix} = \begin{pmatrix} \frac{64}{7} & \frac{74}{7} & 3 \\ -\frac{51}{7} & -\frac{57}{7} & 0 \\ -\frac{53}{7} & -\frac{58}{7} & 0 \end{pmatrix} \cdot \begin{pmatrix} -\frac{16}{9} & \frac{8}{9} & -\frac{1}{9} \\ \frac{14}{9} & -\frac{7}{9} & \frac{2}{9} \\ -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{64}{7}\cdot\left(-\frac{16}{9}\right) + \frac{74}{7}\cdot\frac{14}{9} + 3\cdot\left(-\frac{1}{9}\right) & \frac{64}{7}\cdot\frac{8}{9} + \frac{74}{7}\cdot\left(-\frac{7}{9}\right) + 3\cdot\frac{2}{9} & \frac{64}{7}\cdot\frac{8}{9} + \frac{74}{7}\cdot\left(-\frac{7}{9}\right) + 3\cdot\frac{2}{9} \\ -\frac{51}{7}\cdot\left(-\frac{16}{9}\right) + \left(-\frac{57}{7}\right)\cdot\frac{14}{9} + 0\cdot\left(-\frac{1}{9}\right) & -\frac{51}{7}\cdot\frac{8}{9} + \left(-\frac{57}{7}\right)\cdot\left(-\frac{7}{9}\right) + 0\cdot\frac{2}{9} & -\frac{51}{7}\cdot\frac{8}{9} + \left(-\frac{57}{7}\right)\cdot\left(-\frac{7}{9}\right) + 0\cdot\frac{2}{9} \\ -\frac{53}{7}\cdot\left(-\frac{16}{9}\right) + \left(-\frac{58}{7}\right)\cdot\frac{14}{9} + 0\cdot\left(-\frac{1}{9}\right) & -\frac{53}{7}\cdot\frac{8}{9} + \left(-\frac{58}{7}\right)\cdot\left(-\frac{7}{9}\right) + 0\cdot\frac{2}{9} & -\frac{53}{7}\cdot\frac{8}{9} + \left(-\frac{58}{7}\right)\cdot\left(-\frac{7}{9}\right) + 0\cdot\frac{2}{9} \end{pmatrix}$$

$$\begin{pmatrix} \frac{64}{7}\cdot\left(-\frac{1}{9}\right) + \frac{74}{7}\cdot\frac{2}{9} + 3\cdot\left(-\frac{1}{9}\right) \\ -\frac{51}{7}\cdot\left(-\frac{1}{9}\right) + \left(-\frac{57}{7}\right)\cdot\frac{2}{9} + 0\cdot\left(-\frac{1}{9}\right) \\ -\frac{53}{7}\cdot\left(-\frac{1}{9}\right) + \left(-\frac{58}{7}\right)\cdot\frac{2}{9} + 0\cdot\left(-\frac{1}{9}\right) \end{pmatrix} = \begin{pmatrix} -\frac{1}{7} & \frac{4}{7} & 1 \\ \frac{2}{7} & -\frac{1}{7} & -1 \\ \frac{4}{7} & -\frac{2}{7} & -1 \end{pmatrix}$$